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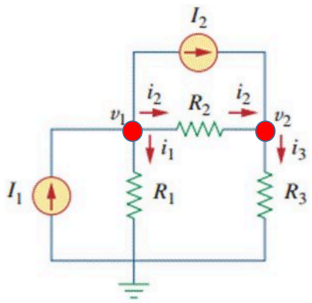
Course no: Phy 4143
Course Title: Physics II
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Professor

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Nodal Analysis

- **Kirchhoff's Current law**
- **Ohm's law**

Nodal Analysis Definition: Nodal analysis is a method for analyzing electric circuits by calculating voltages at different points, or nodes, in the circuit.
Kirchhoff's Current Law (KCL): KCL is used in nodal analysis to ensure the total current entering a node equals the total current leaving, which is fundamental for setting up the equations.
Reference and Non-Reference Nodes: Reference nodes are used as a baseline or zero voltage point, while non-reference nodes are the points where voltage is measured and calculated.
Handling of Components: In nodal analysis, components like resistors and current sources are used to formulate equations based on Ohm's Law.

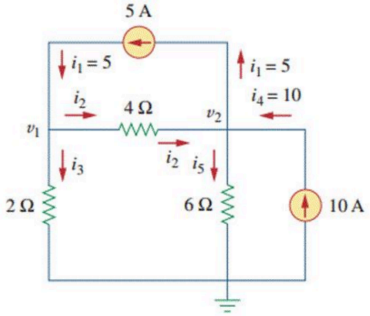


- **Identify the nodes**
- **Apply KCL at each node**
- **Write Equations for each node using Ohm's law**
- **Solve the equation to find Node voltages**

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Nodal Analysis with Current Sources
Nodal analysis with current sources is very easy and it is discussed with an example below.

Example: Calculate Node Voltages in following circuit



In circuit we have 3 nodes from which one is reference node and other two are non reference nodes – Node 1 and Node 2.

Step I. Assign the nodes voltages as v1 and 2 and also mark the directions of branch currents with respect to the reference nodes

Step II. Apply KCL to Nodes 1 and 2
KCL at Node 1

$$i_2 + i_4 = i_1 + i_5 \cdots\cdots (2) \Rightarrow i_1 = i_2 + i_3 \cdots\cdots (1)$$

Step III. Apply Ohm's Law to KCL equations Node 1 & 2 and simplifying

$$i_1 = i_2 + i_3 \Rightarrow 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2} \Rightarrow 3v_1 - v_2 = 20 \cdots\cdots (3)$$

$$i_2 + i_4 = i_1 + i_5 \Rightarrow \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6} \Rightarrow -3v_1 + 5v_2 = 60 \cdots\cdots (4)$$

Step IV. Solving (3) and (4) gives
V_1 = 13.33 Volts and V_2 = 20 Volts

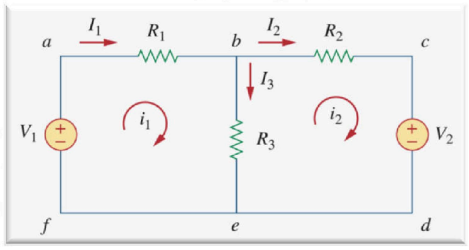
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Mesh Analysis → • Kirchhoff's Voltage law • Ohm's law

- Mesh analysis is also known as loop analysis or the mesh current method.
- Nodal analysis applies KCL to find unknown voltages in a given circuit, while mesh analysis applies KVL to find unknown currents.
- The current through a mesh is known as mesh current.

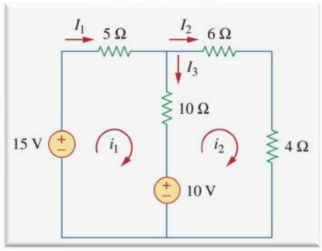
Steps

1. Make a clear diagram.
2. Assign mesh currents $i_1, i_2, i_3, \ldots, i_n$ to the n meshes.
3. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
4. Solve the resulting n simultaneous equations to get the mesh currents.



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Example 1: For the shown circuit, find the branch currents and using mesh analysis.



We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$3i_1 - 2i_2 = 1 \qquad (1)$$

For mesh 2

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$i_1 - 2i_2 = -1 \qquad (2)$$

Solve equations 1 and 2 to get i_1 and i_2.

$$3i_1 - 2i_2 = 1 \qquad (1)$$

$$i_1 - 2i_2 = -1 \qquad (2)$$

$$2i_1 = 2$$

$$\boxed{i_1 = 1\text{A}}$$

Substitute i_1 in eq.1 or eq.2 to get i_2

$$3 \times 1 - 2i_2 = 1$$

$$-2i_2 = 1 - 3 = -2$$

$$\boxed{i_2 = 1\text{A}}$$

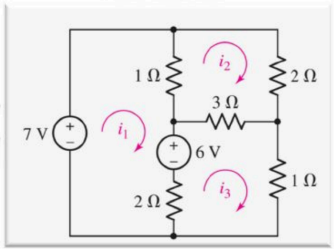
$$\boxed{I_1 = i_1 = 1\text{A}} \qquad \boxed{I_2 = i_2 = 1\text{A}}$$

$$\boxed{I_1 = i_1 - i_2 = 1 - 1 = 0\text{A}}$$

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Example 2: For the shown circuit, find the mesh currents using mesh analysis.



$$-7 + 1(i_1 - i_2) + 6 + 2(i_1 - i_3) = 0$$

$$1(i_2 - i_1) + 2i_2 + 3(i_2 - i_3) = 0$$

$$2(i_3 - i_1) - 6 + 3(i_3 - i_2) + 1i_3 = 0$$

Simplifying,

$$3i_1 - i_2 - 2i_3 = 1 \qquad (1)$$

$$-i_1 + 6i_2 - 3i_3 = 0 \qquad (2)$$

$$-2i_1 - 3i_2 + 6i_3 = 6 \qquad (3)$$

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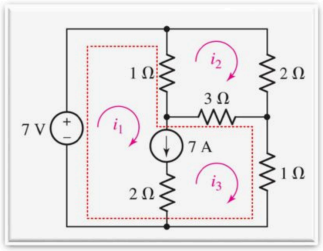
Mesh Analysis with Current Sources

No, thanks

Note the following properties of a supermesh:

1. The current source in the supermesh provides the constraint equation necessary to solve for the mesh currents.
2. A supermesh has no current of its own.
3. A supermesh requires the application of both KVL and KCL.

Example: Determine the three mesh currents in the shown figure.



7A independent current source is in the common boundary of two meshes, which leads us to create a supermesh whose interior is that of meshes 1 and 3. Applying KVL about this loop.

$-7+1(i_1 - i_2)+3(i_3 - i_2)+1i_3=0$

$i_1 - 4i_2+4i_3=7$ (1)

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For mesh 2

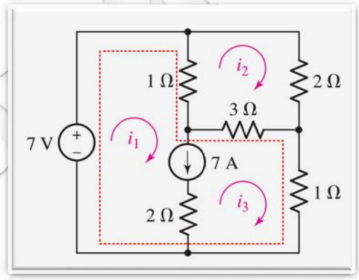
$1(i_2 - i_1)+2i_2+3(i_2 - i_3)=0$

$-i_1 + 6i_2 - 3i_3=0$ (2)

the independent source current is related to the mesh currents

$i_1 - i_3=7$ (3)

Solve these three equation to get mesh currents

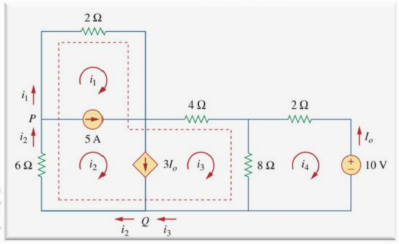


$i_1=9A$ $i_2=2.5A$ $i_3=2A$

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Mesh Analysis with Current Sources

Example 4: For the shown circuit, find i_1 to i_4 using mesh analysis.



Note that meshes 1 and 2 form a supermesh since they have an independent current source in common. Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common. The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh

$2i_1+4i_3+8(i_3 - i_4)+6i_2=0$

$2i_1 + 6i_2 + 12i_3 - 8i_4=0$ (1)

For the independent current source, we apply KCL to node P:

$i_2 = i_1 +5$ (2)